



Mapping antibiotic use and calf-rearing practices on Brazilian dairy farms

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ABSTRACT

Antimicrobial use in preweaning dairy calves is largely driven by the high morbidity and mortality observed in early life and by on-farm management practices. However, such use also contributes to the emergence and dissemination of antimicrobial-resistant bacteria, raising important public and environmental health concerns. This study aimed to characterize antimicrobial use patterns and management practices in Brazilian dairy herds. An online questionnaire was distributed between June and November 2020 via social media, e-mail, and WhatsApp messages to reach dairy farm managers, including producers, farm owners, and professional consultants. After data curation, statistical analyses were performed using multiple correspondence analysis followed by cluster analysis to identify production system typologies. Based on data from 725 dairy farms, 4 clusters were identified, representing a continuum from intensive to low-input systems. Cluster 3 ($n = 101$) consisted of large, well-structured farms with standardized rearing protocols, regular veterinary oversight, and more judicious antimicrobial use based on clinical signs, although critically important antimicrobials for human medicine were still frequently used. Clusters 1 ($n = 205$) and 2 ($n = 335$) represented intermediate systems and included most farms; cluster 1 featured moderately consistent management practices, whereas cluster 2 was characterized by empirical, unsupervised antimicrobial use. Cluster 4 ($n = 84$) encompassed smallholder farms relying on natural suckling and older-generation antimicrobials such as penicillin. Waste milk feeding was widespread across clusters, particularly in clusters 1 and 2, and was often improperly discarded into sewage systems. In summary, these results reveal substantial het-

erogeneity in antimicrobial use and management among Brazilian dairy systems. They highlight the influence of veterinary supervision and management quality on antimicrobial decision-making, while emphasizing that larger farms with regular veterinary assistance should consider the World Health Organization's list of medically important antimicrobials to avoid the use of drugs prioritized for human medicine. This study provides an epidemiological baseline to support targeted antimicrobial stewardship and educational strategies tailored to diverse production contexts.

Key words: diarrhea, bovine respiratory disease, waste milk, antibiotic

INTRODUCTION

The preweaning period is a critical stage in the life of dairy calves, characterized by physiological immaturity, nutritional transitions, and heightened disease susceptibility. Management practices during this phase, such as colostrum administration, feeding protocols, and housing conditions, directly affect calf survival, growth, and long-term productivity. Diarrhea and bovine respiratory disease (BRD) are the leading causes of morbidity and mortality during this period. Diarrhea can affect up to 100% of calves in the first 3 wk of life, and BRD affects 12% to 20% of preweaning calves (Windeyer et al., 2014; USDA, 2018; Azevedo et al., 2020). To manage these health threats, dairy producers frequently use antimicrobials for both therapeutic and metaphylactic purposes.

Despite guidelines recommending antimicrobial use only for confirmed or suspected bacterial infections, broad-spectrum and critically important drugs are often used without diagnostic confirmation or susceptibility testing (WHO, 2024). In countries like Brazil, their widespread availability and the absence of mandatory veterinary prescriptions contribute to inappropriate use, further exacerbated by inconsistent veterinary

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

access (Tomazi and dos Santos, 2020). For diarrhea, sulfonamides, amoxicillin, and ampicillin are generally recommended as first-line agents, and fluoroquinolones are discouraged (Constable, 2009; Constable et al., 2017). Bovine respiratory disease is typically treated with florfenicol, tulathromycin, and other macrolides (Lubbers and Turnidge, 2015; O'Connor et al., 2016; Hubbuch et al., 2021). In addition, the use of waste milk (WM) represents an indirect exposure to pathogenic bacteria and subtherapeutic antibiotic levels (Brunton et al., 2012; Maynou et al., 2017; Tempini et al., 2018; Ma et al., 2022).

Antimicrobial resistance is a growing global concern, and calves have been identified as important reservoirs of resistant bacteria within dairy systems (Duse et al., 2015a). Calves aged 2–3 wk are particularly at risk because of a combination of immunological vulnerability, high disease incidence, and frequent exposure to antimicrobials (Berge et al., 2010; Aust et al., 2013; Springer et al., 2019). In addition, the use of antimicrobials as a preventive strategy has been shown to be largely ineffective in reducing disease incidence and is associated with long-term effects on gut microbiota development and resistance gene dissemination (Zhang et al., 2022; Cangiano et al., 2023; Garzon et al., 2024).

Brazil is among the world's 5 largest milk producers, with cow milk production totalling approximately 24.7 million tonnes (24.7 billion liters) in 2023 (USDA, 2025). The diversity of production systems, regional disparities, and uneven access to veterinary support present unique challenges and opportunities for improving calf health and promoting antimicrobial stewardship. However, comprehensive data on calf management, antimicrobial use, and WM feeding practices remain limited, particularly among small- and medium-scale dairy farms.

This study aimed to characterize the management of preweaning dairy calves in Brazil, focusing on antimicrobial use and the practices involving of WM as a liquid diet. We hypothesized that regional factors, including geographic, climatic, and socioeconomic conditions across Brazilian states, shape calf-rearing practices and influence antimicrobial use. Specifically, we expected that farms exposed to less favorable conditions would display greater reliance on antimicrobials, reflecting increased disease pressure and limited veterinary oversight. Conversely, we anticipated that larger and better-structured farms would demonstrate more judicious antimicrobial use, greater veterinary involvement, and more regulated WM management. By identifying management typologies associated with antimicrobial use patterns, this study provides a foundation for the design of targeted health programs and evidence-based antimicrobial stewardship strategies within the One Health framework.

MATERIALS AND METHODS

Questionnaire Development

This cross-sectional study was conducted using a questionnaire designed to assess management practices and the use of antimicrobials and WM for milk-fed preweaning dairy calves in Brazil.

The questionnaire was developed based on the following bibliographic references: Brunton et al. (2012), Duse et al. (2013), Randall et al. (2014), Duse et al. (2015a), Duse et al. (2015b), Klein-Jöbstl et al. (2015), Higham et al. (2018), and Tempini et al. (2018).

The questionnaire consisted of the general property information – property features (6 variables) and a total of 59 questions distributed into 5 thematic sections: use of antimicrobials in lactating and dry cows (questions 1–16); colostrum management and calf care (questions 17–30); feeding calves during the preweaning phase (questions 31–37); use of antimicrobials for preweaning calves (questions 38–51); and use of WM for preweaning calves (questions 52–59). The variables analyzed in the present paper included general farm information and sections 2 to 5 (colostrum management and calf care through the use of WM for preweaning calves), which were specific to preweaning calves. After data curation, a subset of questions with complete and consistent responses was used for further statistical analyses, as described later. The full questionnaire is available as supplemental material (see Notes).

Data Collection Method and Criteria for Excluding Questionnaires and Responses

All data collection procedures, including the collection of farm-related information, were approved by the National Research Ethics Commission (CONEP) and the institutional Research Ethics Committees (CEP) under Plataforma Brasil (protocol no. 359666620.2.0000.5422), as well as by the Ethics Committee on the Use of Animals (protocol no. 4964240720).

Farms were recruited through convenience and snowball sampling based on voluntary participation. The questionnaire link was disseminated nationally through professional networks, dairy cooperatives, producer associations, and veterinary extension services. Although the survey was not based on probabilistic sampling, the responses represented a wide geographic distribution across all major milk-producing regions of Brazil and included farms of different herd sizes, management levels, and production intensities. Therefore, although the sample cannot be considered statistically representative of the entire Brazilian dairy industry, it captures the diversity of production systems and management practices,

providing valuable insights into antimicrobial use patterns and typologies that reflect real-world conditions.

The survey was designed using Google Forms (www.google.com/forms), ensuring compatibility with both desktop and mobile devices. The Google Forms link was distributed via social media platforms (Instagram, Facebook, and WhatsApp) to facilitate broad and low-cost access, particularly given the restrictions on in-person data collection during the COVID-19 pandemic.

The target audience included individuals directly involved in dairy farm operations such as farm owners, staff responsible for animal management, and veterinarians providing technical assistance. These groups were identified by analyzing the social media profile, in addition to the professional dairy networks, regional producer associations, and dairy industries partnership. The snowball recruitment strategy allowed the survey to reach farms through these existing contacts. Although the number of respondents from each group varied, all groups were represented, and the farm was considered the primary unit of analysis rather than the individual respondent.

Responses were monitored for completeness and internal consistency to ensure data reliability. Participation was entirely voluntary and anonymous. No personal identifiers (e.g., names, contact details, or farm geolocation data) were collected, and responses were stored in a secure, password-protected database accessible only to the research team. Each participant provided electronic informed consent before accessing the questionnaire, which described the study objectives, confidentiality policy, and voluntary nature of participation. Although participants could share the survey link with others, no referral tracking or participant identification was possible, thereby minimizing the risk of inadvertent disclosure of third-party information.

Before the actual survey, the questionnaire was pilot tested to assess the need for adjustments, clarity, and the estimated time required for respondents to complete the entire form. The pilot phase was conducted by the research technical team under the supervision of the postgraduate student responsible for the study and her advisor. The questionnaire was available online for 1 mo, from December 2019 to January 2020, and 106 responses were received. These responses were individually reviewed to identify comprehension issues and incomplete or inconsistent items. Based on this evaluation, 4 questions related to general farm information and property characteristics were removed due to low response rates or ambiguous wording. In addition, minor rewording and adjustments in question order were made to improve flow and readability. No new questions were added. After these refinements, the questionnaire was finalized for national data collection.

The final version of the questionnaire consisted of 59 questions distributed into 5 thematic sections. However, 16 questions related to antimicrobial use in lactating and dry cows were excluded from the present statistical analysis because these data will be reported separately in part 2 of this study. Therefore, 43 questions focused on preweaning calf management were retained for analysis. In addition, 6 non-nominal variables describing general property information were included, resulting in a total of 49 variables organized during the data curation phase and used for subsequent statistical analyses. The final Google Forms questionnaire was available online for 5 mo, from June to November 2020.

To reduce respondent fatigue and prevent incomplete submissions in the 59-question survey, questions were organized into short, thematic sections with simple, closed-ended, or multiple-choice formats. Most questions required selecting predefined options, with "Other" fields allowing brief clarifications. The questionnaire was pretested to ensure clarity and conciseness, and the average completion time was approximately 12 to 15 min, which was considered acceptable for producers and veterinarians. The online format allowed participants to complete the form at their convenience and resume later if necessary. Responses were reviewed for completeness and internal consistency during the data curation phase. Options such as "Not applicable," "No opinion," or "Other" were treated as missing data in statistical analyses and excluded from the multiple correspondence analysis (MCA) to avoid bias. The proportion of such responses was minimal (<5%) and did not affect the representativeness or robustness of the results.

All responses were reviewed during the data curation phase to ensure completeness, logical consistency, and accuracy before statistical analysis. The curation was conducted by the postgraduate student responsible for the project, under the direct supervision of her advisor. Each dataset (farm record) was examined individually to identify incomplete, inconsistent, or duplicated responses. A standardized verification protocol was applied, including predefined criteria for data exclusion (e.g., missing key variables or contradictory answers). When discrepancies were identified, data were cross-checked and validated jointly by the postgraduate student and her advisor before any exclusion or correction. This standardized approach ensured uniformity, reproducibility, and reliability of the final dataset used for statistical analyses.

Statistical Analysis

A total of 1,082 questionnaires were completed; however, 48 were excluded according to predefined criteria, resulting in 1,034 valid responses for analysis. Questionnaires were excluded if they were incomplete

Table 1. Descriptive presentation of the 725 dairy farms included in the calf management and antimicrobial use survey

Statistic	Number of lactating cows	Milk production (L/d)	Average milk yield per cow (L/d)	Total cattle inventory (head)
Minimum	3	20	3.3	7
Lower quartile (Q1)	20	360	16.1	45
Median	37	740	21.9	85
Upper quartile (Q3)	77	1,800	28.0	164
Maximum	2,040	81,000	46.2	4,720
Mean $\pm$ SD	85.0 $\pm$ 175.2	2,491.5 $\pm$ 6,463.8	22.1 $\pm$ 8.3	187.3 $\pm$ 372.2
CV (%)	206.2	259.4	37.5	198.7

(more than 20% missing data), duplicated, or presented inconsistent information across sections. These criteria were defined a priori to ensure data quality and consistency, and their application did not affect the overall distribution of the dataset.

For the purposes of multivariate statistical analysis, only farms with complete data for all 49 categorical variables were retained, resulting in a final sample of 725 farms. This reduction was necessary due to the requirements of the MCA method, which does not accommodate missing values and relies on complete response sets to generate reliable dimensions that represent the underlying structure of the data.

The multiple variables were analyzed using MCA, a technique that reduces the dimensionality of data, similar to principal component analysis (PCA). However, while PCA is applied to continuous or numerical variables, MCA is specifically used for categorical variables, resulting in new variables known as dimensions. Multiple correspondence analysis uses the inertia statistic, which is analogous to the eigenvalue in PCA.

Based on the inertia values adjusted using the correction of Benzécri (1979), the first 2 dimensions were retained, representing the inflection point of the scree plot and jointly explaining 72% of the total variance. Retaining additional dimensions beyond the second did not substantially increase the explained inertia or improve interpretability; therefore, 2 dimensions were considered sufficient to capture the major underlying structure of the dataset. This approach ensured a parsimonious and stable representation for subsequent cluster analysis.

In the sequence, a cluster analysis was conducted to group farms based on those 2 retained dimensions, which capture the information from the 49 original question variables. A nonhierarchical clustering method was applied based on Euclidean distances, where cluster centers were estimated using the least-squares criterion. This approach resulted in the identification of 4 distinct farm clusters.

Subsequently, new MCA were conducted using the original categorical variables, but also incorporating the clusters as an additional variable. Variables were grouped based on biological relationships to facilitate interpreta-

tion of the cluster profiles. Finally, MCA charts were created to explore the associations between the question variables and the clusters. Some continuous variables, such as number of lactating cows, daily mean milk yield per cow and per farm, were compared between the clusters using a general linear model with Tukey–Kramer post hoc test for pairwise comparisons.

All the statistical analyses were performed in SAS University Edition (SAS Institute Inc., 2022) considering the level of 5% of significance. The MCA was performed using SAS PROC CORRESP, while dairy farm clustering was conducted with SAS PROC FASTCLUS. The graphical representation of the farm groupings was generated using SAS PROC CANDISC. Frequencies of responses to the questionnaire were calculated using SAS PROC FREQ, and general linear modeling was performed using SAS PROC GLIMMIX.

RESULTS

Dataset Overview

The 725 dairy farms included in the survey showed wide variation in herd size and productivity (Table 1). Most dairy producers were located in Brazil's south and southeast regions, particularly in Rio Grande do Sul, Santa Catarina, Paraná, Minas Gerais, and São Paulo, where dairy farming is most concentrated. Holstein was the predominant breed across herds in these regions.

The number of lactating cows ranged from 3 to 2,040, with a median of 37 cows and a mean ($\pm$ SE) of 85 $\pm$ 175.2, reflecting the coexistence of smallholders and large-scale operations within the sample. Daily milk production varied between 20 and 81,000 L, with a median of 740 L and a mean of 2,491.5 $\pm$ 6,463.8 L, and the average milk yield per cow was 22.1 $\pm$ 8.3 L/d (range: 3.3–46.2 L/d). Total cattle inventory ranged from 7 to 4,720 animals, with a median of 85 and a mean of 187.3 $\pm$ 372.2. The high CV for all variables underscore the structural and productive heterogeneity of Brazilian dairy systems. Monthly veterinary assistance was reported in 39.5% of farms, whereas 29.0% had no regular visits.

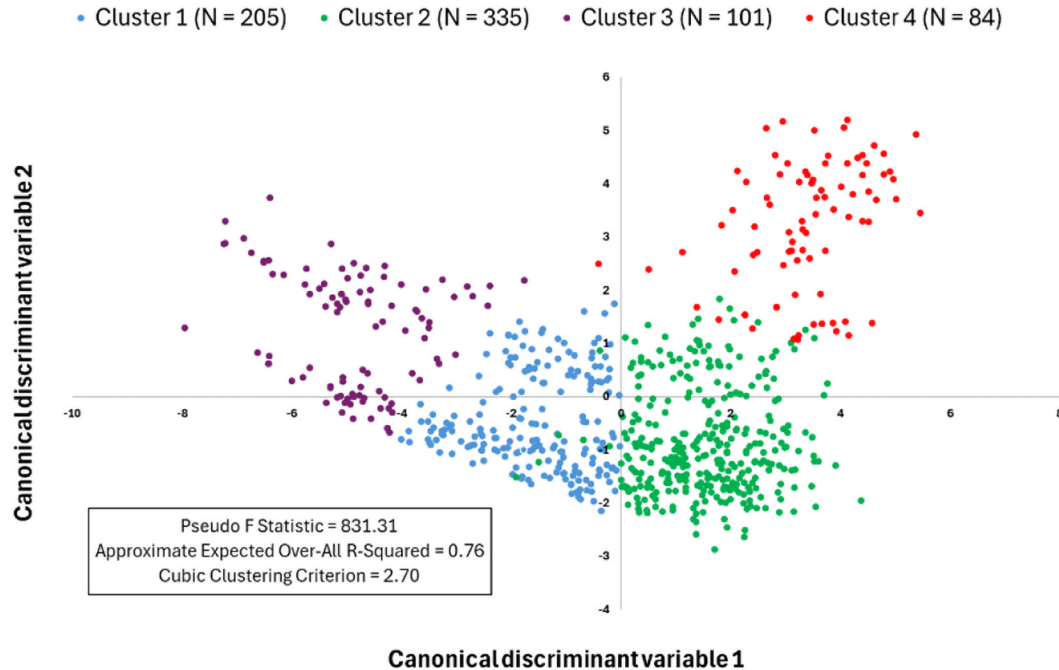


Figure 1. Canonical discriminant analysis of 725 Brazilian dairy farms based on 44 dairy farm practices variables.

Colostrum and early feeding management showed substantial room for improvement. Only 24.1% of farms assessed colostrum quality, and natural suckling was the main feeding method (48.6%), followed by nipple bottle (45.8%). Colostrum replacers were rarely used. Feeding volumes were typically ad libitum (36.8%) or ≤ 3 L (34.3%), and the timing of first feeding was often delayed: 24.1% reported natural suckling and 19.9% provided colostrum 2 h after birth. A second feeding was uncommon (absent in 87.9% of farms). Colostrum banks were reported by 42.8% of farms, but pasteurization was rare (2.6%). Most producers used colostrum from cows treated during the dry period. Transition milk was routinely offered in 54.6% of farms, and natural suckling persisted in 9.5%.

Liquid feeding practices were diverse: saleable milk (45.9%), WM (19.6%), mixed sources (18.8%), milk replacer (7.6%), and natural suckling (8.1%). Only 30.3% of farms adopted intensive feeding (>3 L per meal), and 89.8% fed twice daily. Individual pens predominated for preweaning calves, and collective feeding was uncommon (4.8%). Buckets were the most frequent feeding equipment.

Waste milk management was common: 44% of farms fed WM regularly and 18.5% occasionally, mainly to reduce costs (29.8%), for convenience (10.1%), or due to disposal difficulties (6.9%). In 30.3% of farms, all WM was fed to calves; 52.1% discarded it in sewage, and 13.7% used it for other species. Waste milk was typically introduced during the first week of life.

Regarding antimicrobial use and decision-making, veterinarians provided treatment guidance in only 17.4% of farms. Diarrhea was the main indication for antimicrobial therapy, followed by BRD. For diarrhea, the most used antimicrobial classes were sulfonamides, quinolones, tetracyclines, and β -lactams, sulfadiazine being the most common active ingredient (27.5%). For BRD, β -lactams (23.2%), quinolones (20.8%), penicillin (20.0%), and amphenicols (13.5%) predominated. Notably, all calves diagnosed with diarrhea received antimicrobials (51.6%), whereas fever guided treatment decisions in only 14.2% of cases. Preventive or metaphylactic antimicrobial use was reported in 22.5% of farms (routine) and 10.9% (occasional), most often using β -lactams, tetracyclines, macrolides, quinolones, sulfonamides, and amphenicols, typically administered during the first week of life.

Detailed descriptive statistics are provided in Supplemental Tables S1 to S4 and Figures S1 to S10 (see Notes).

Canonical Discriminant Analysis of Cluster Separation

The canonical discriminant analysis (CDA) showed a clear and statistically robust separation among the 4 clusters (Figure 1). The first 2 canonical discriminant functions explained 76% of the total variance, with high model discrimination (pseudo $F = 831.3$) and a positive cubic clustering criterion of 2.70. This separation confirmed the consistency of the clustering procedure and

Table 2. Descriptive profiles of the 4 clusters identified among 725 Brazilian dairy farms

Variable	Cluster 1 (n = 205)	Cluster 2 (n = 335)	Cluster 3 (n = 101)	Cluster 4 (n = 84)	P-value
Total cattle on the farm, n	143 ± 8.2 ^b	57 ± 2.6 ^c	343 ± 28.0 ^a	49 ± 4.4 ^c	<0.001
Lactating cows, n	65 ± 3 ^b	25 ± 1.1 ^c	169 ± 13.6 ^a	15 ± 1.4 ^d	<0.001
Daily milk production, L	1,689 ± 110.9 ^b	458 ± 23.5 ^c	5,199 ± 486.9 ^a	172 ± 17.7 ^d	<0.001
Average milk yield per cow, L	26.5 ± 0.42 ^b	19.0 ± 0.33 ^c	31.3 ± 0.59 ^a	12.2 ± 0.65 ^d	<0.001

^{a-d}Within rows, means with different superscripts differ among clusters ($P < 0.05$).

supported the interpretation of the 4 clusters as distinct production typologies, representing gradients in herd size, management intensity, and veterinary oversight.

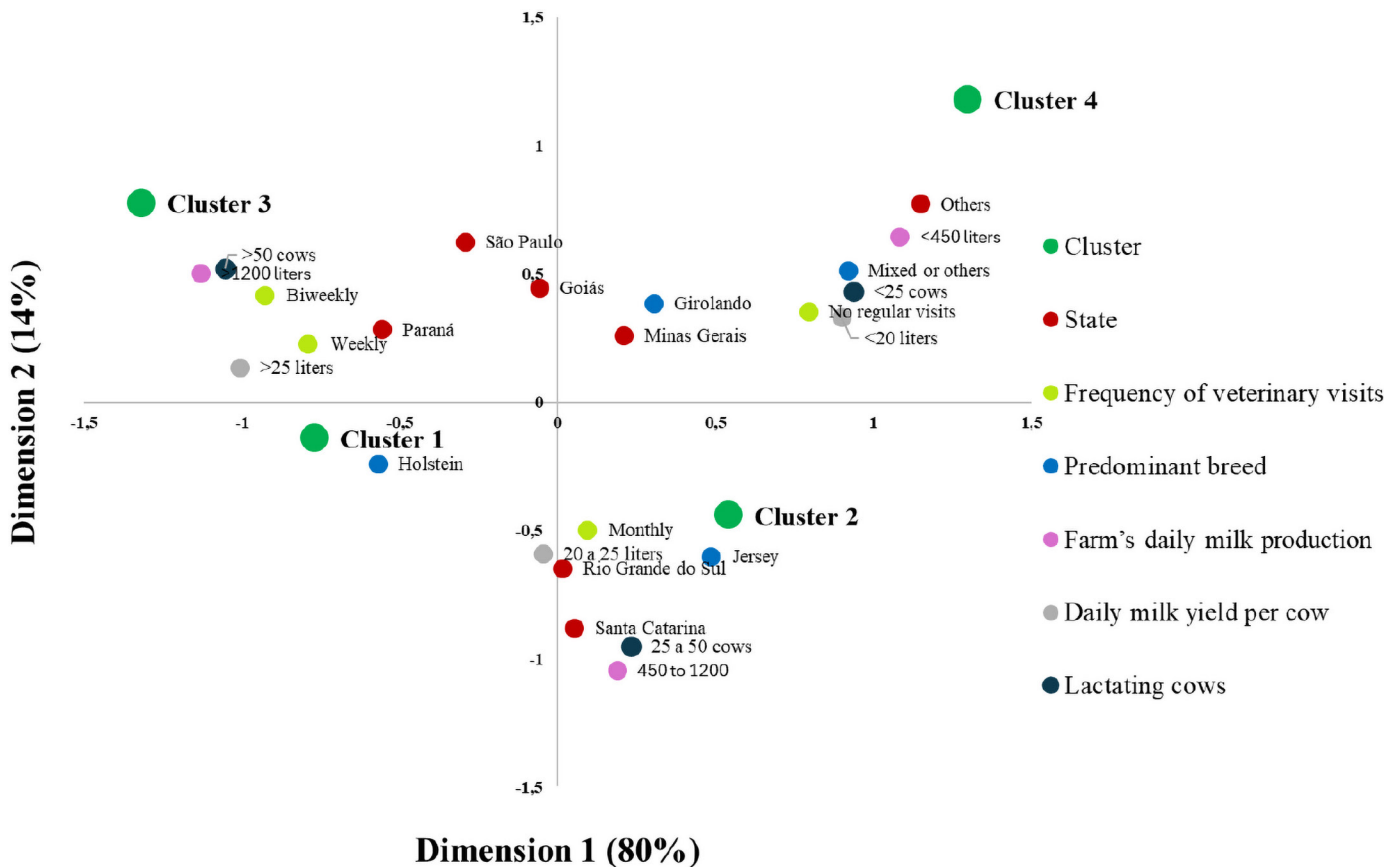
Characterization of Dairy Farm Clusters

Descriptive statistics of herd characteristics across the 4 clusters are summarized in Table 2. Significant differences were observed among clusters for all variables analyzed ($P < 0.001$). Cluster 3 represented the most intensive production system, with the largest herds and highest productivity. Cluster 1 showed intermediate-to-high performance, whereas clusters 2 and 4 encompassed smaller, low-yield herds typical of family-based or low-input systems. These

gradients clearly distinguished production intensification levels among Brazilian dairy farms and supported the typologies identified through cluster analysis.

Multivariate Characterization of Herd-Level Profiles

The MCA explored associations among herd-level categorical variables and the 4 clusters identified by the CDA (Figure 2). The first 2 dimensions explained 94% of the total variance (80% and 14% for dimensions 1 and 2, respectively), indicating a strong structure in the dataset. Dimension 1 primarily reflected a gradient of production intensity and management quality. Clusters 3 and 1 were associated with larger, higher-yield farms reporting

**Figure 2.** Multiple correspondence analysis of herd-level characteristics and cluster classification in Brazilian dairy farms.

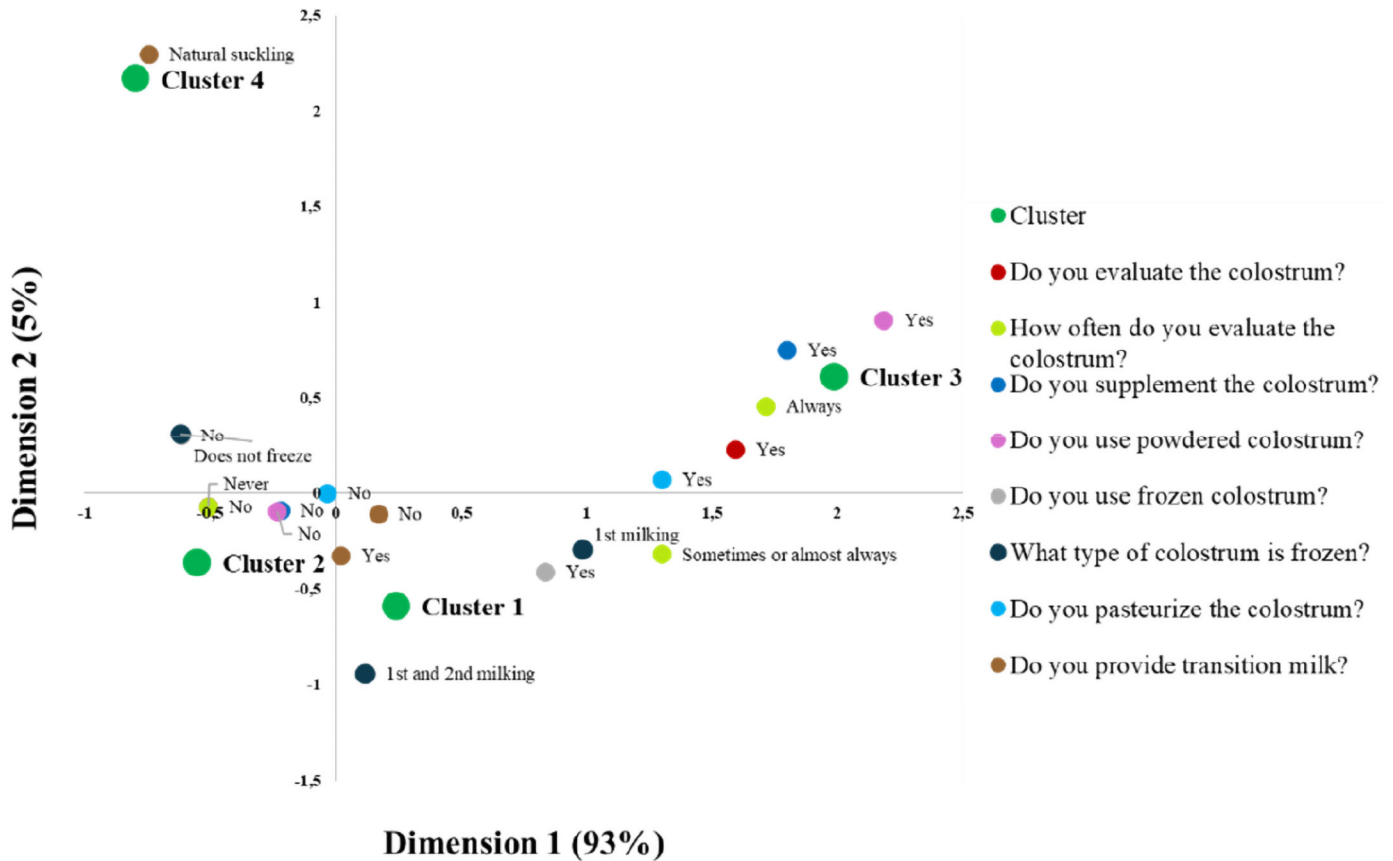


Figure 3. Multiple correspondence analysis of colostrum source and management practices in relation to cluster classification.

regular veterinary oversight, predominance of specialized dairy breeds, and location in major dairy states such as São Paulo and Paraná. In contrast, Clusters 2 and 4 represented smaller herds with lower productivity and limited technical assistance, typically located in southern or less specialized regions. Overall, the MCA confirmed that the clusters represented distinct production typologies, ranging from intensive, well-managed herds to low-input, family-based systems with limited veterinary and technical support.

Colostrum Management Practices

The MCA examining colostrum source and quality explained 98% of the total variance (93% and 5% for dimensions 1 and 2, respectively; Figure 3). Clusters 3 and 1 were associated with structured colostrum programs, including the use of first-milking colostrum, routine quality assessment, and availability of stored or pasteurized high-quality colostrum. These farms demonstrated adherence to gold-standard practices, although transition milk feeding was not always provided. In contrast, clusters 2 and 4, composed mainly of smaller or family-operated farms, relied more on natural suckling

and showed limited adoption of controlled colostrum handling. Interestingly, these farms were more likely to feed transition milk in the first days of life.

The MCA assessing colostrum feeding practices explained 95% of the total variance (75% and 20% for dimensions 1 and 2, respectively; Figure 4). Clusters 3 and 1 were associated with recommended feeding practices, including timely colostrum delivery within 2 h after birth and adequate volumes equivalent to 10%–15% of BW. These farms predominantly used bottles or orogastric tubes and avoided colostrum from dry-treated cows. However, cluster 1 exhibited greater variability in the consistency of second feedings and delivery methods compared with cluster 3. In contrast, clusters 2 and 4 were characterized by delayed or uncontrolled feeding, low colostrum volumes, and reliance on natural suckling or bucket feeding. Cluster 4, in particular, reflected low-intervention systems, where calves had direct access to dams.

Calf Feeding Management

The MCA exploring associations between calf feeding and housing practices explained 95% of the total vari-



nomically driven systems that maximized the use of available WM from early life onward, and cluster 2 included farms using it primarily for convenience or due to disposal difficulties. In both cases, WM was introduced early in life and constituted a major component of the liquid diet. Conversely, clusters 3 and 4 showed limited or no use of WM. Larger farms in cluster 3 tended to discard milk via sewage systems, whereas smaller farms in cluster 4 commonly redirected it to other animals or pastures.

The Use of WM

The MCA evaluating the association between farm clusters and disease indications for antimicrobial treatment explained 93% of the total variance (66% and 27% for dimensions 1 and 2, respectively; Figure 7). Cluster 3 was strongly associated with antimicrobial use for BRD, suggesting that respiratory infections were the main indication for treatment in these more intensive systems. Clusters 1 and 2 were primarily linked to treatments for diarrhea and, to a lesser extent, for tick fever and arthritis condition. In contrast, cluster 4 encompassed farms that

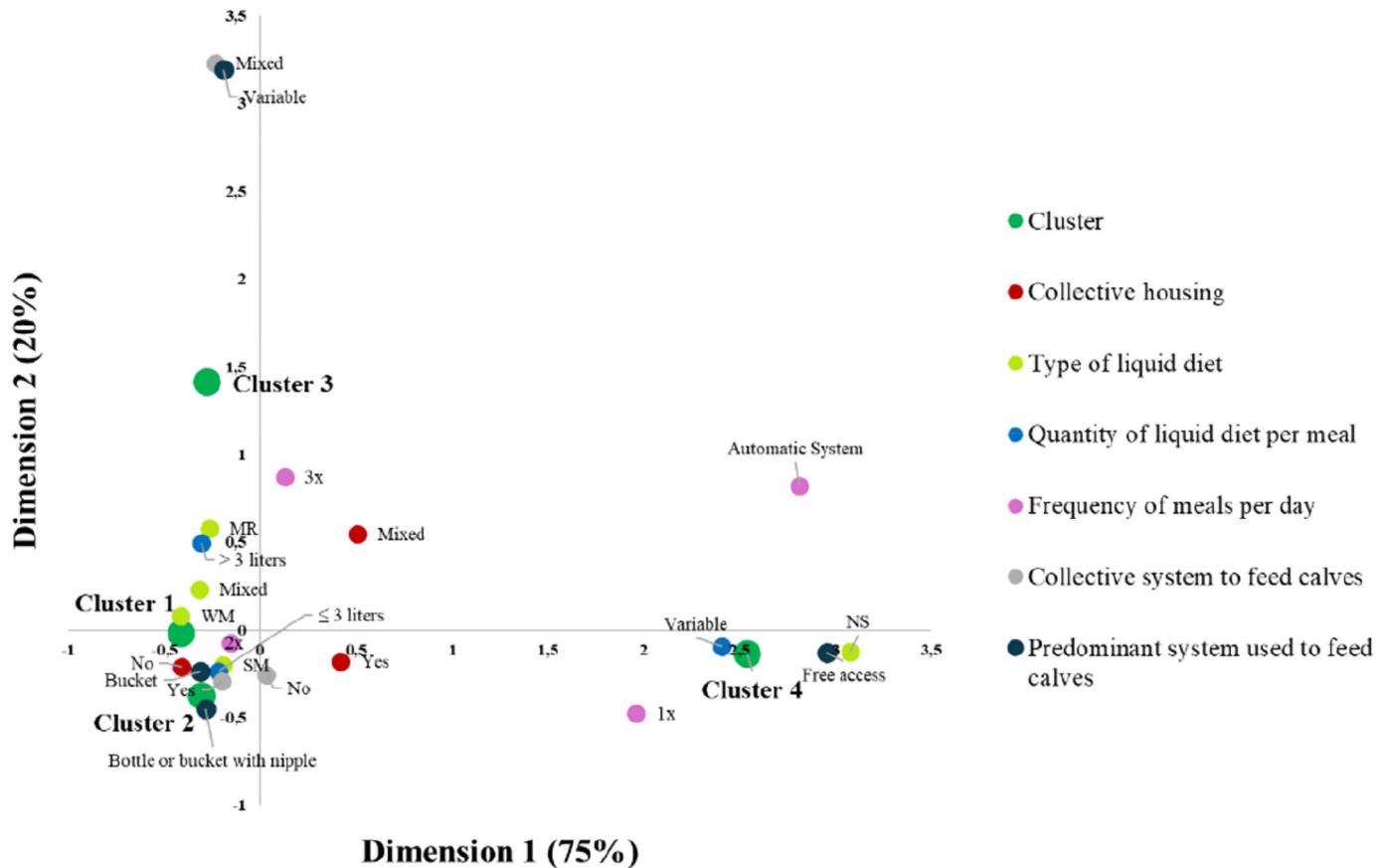


Figure 5. Multiple correspondence analysis of liquid feeding management practices in preweaning calves across farm clusters. MR = milk replacer.

reported minimal or no antibiotic use in calves, except for isolated cases of navel inflammation.

Antibiotic Use for Calves' Diarrhea Therapy

The MCA examining antimicrobial choices and treatment criteria for calf diarrhea explained 54% of the total variance (32% and 22% for dimensions 1 and 2, respectively; Figure 8). Clusters 1 and 3 were associated with more empirical or reactive therapeutic approaches. Cluster 3 tended to rely on sulfonamides, particularly sulfadiazine, and considered fever a key treatment trigger. Cluster 1 showed a pattern of generalized antimicrobial use, often treating all diarrheic calves regardless of severity, and variable decision criteria, including chronic or nonresponsive cases. Cluster 2 represented a broader spectrum treatment profile, with frequent use of β -lactams, tetracyclines, penicillin, and quinolones, reflecting a more intensive or less selective antimicrobial strategy. Cluster 4 showed inconsistent responses, including uncertainty regarding drug choice or use of unspecified active ingredients.

Antibiotic Use for Calves' BRD Therapy

The MCA exploring antibiotic classes and active ingredients used for BRD treatment explained 50% of the total variance (29% and 21% for dimensions 1 and 2, respectively; Figure 9). Distinct antimicrobial use patterns were evident among the 4 clusters. Clusters 1 and 3 were characterized by the use of quinolones (particularly enrofloxacin) and, in the case of cluster 3, additional reliance on amphenicols such as florfenicol. These profiles suggest a tendency toward broad-spectrum antimicrobial selection in more intensive systems. Cluster 2 exhibited no clear association with specific drugs or classes, possibly reflecting low treatment frequency or inconsistent reporting among farms. Cluster 4, which included farms of varying sizes, showed the greatest diversity of antimicrobial use, encompassing β -lactams, tetracyclines, ceftiofur, amoxicillin, and sulfonamides, along with reports of uncertain or unspecified drug selection. The isolated position of penicillin indicates limited use for BRD treatment in preweaning calves.

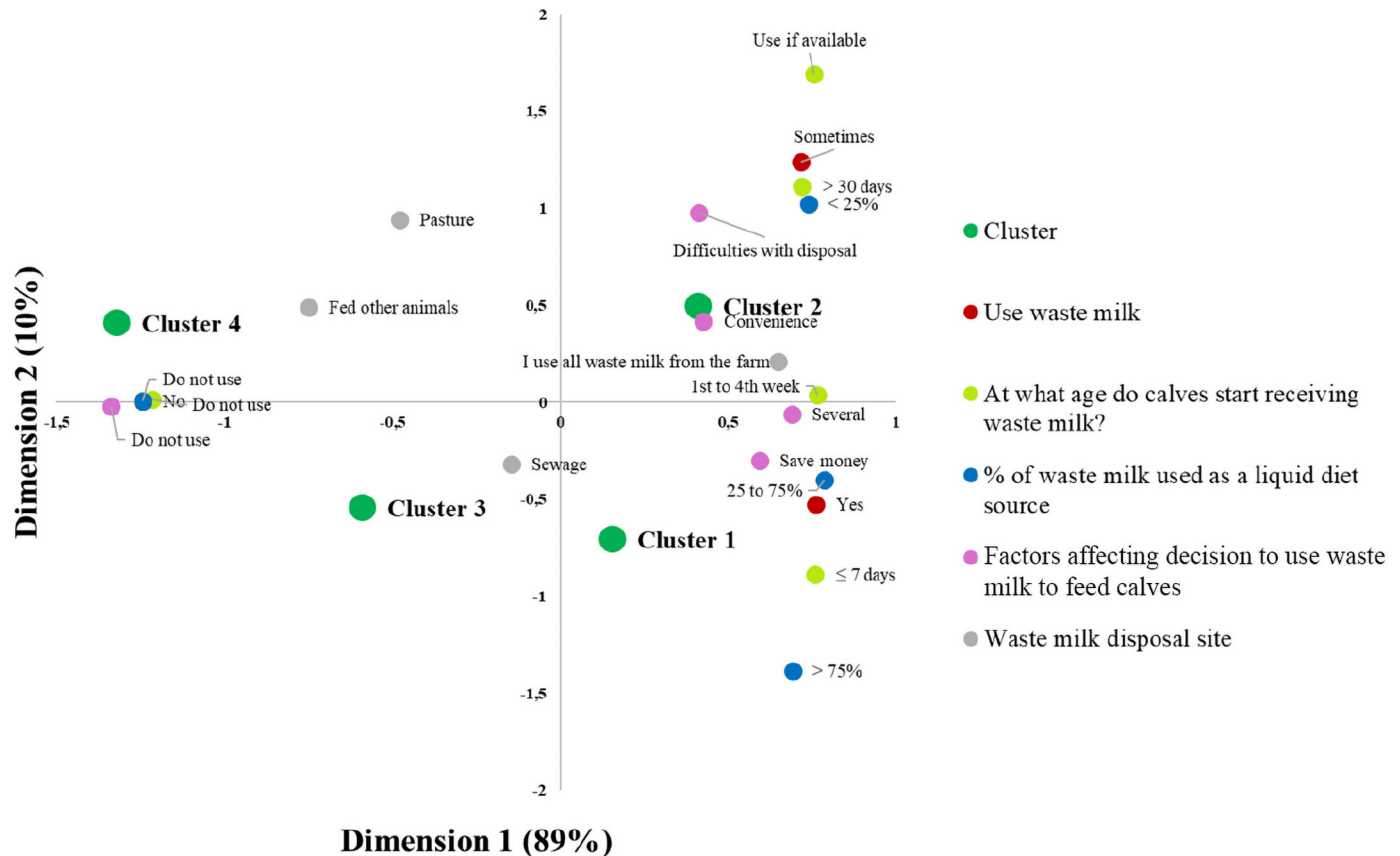


Figure 6. Multiple correspondence analysis of WM use and motivations for feeding practices in Brazilian dairy farms.

Antibiotic Use for Prevention and Metaphylaxis

The MCA exploring antimicrobial use for disease prevention or metaphylaxis explained 97% (Figure 10A) and 91% (Figure 10B) of the total variance in their respective models, revealing consistent patterns across farm clusters. Most farms in cluster 3 reported no antimicrobial use for preventive or metaphylactic purposes, indicating a restrictive or avoidance-based approach to prophylaxis. In contrast, cluster 4 was distinctly associated with veterinarian-guided antimicrobial recommendations, as well as the use of injectables administered during the first days of life. These practices included β -lactams, sulfonamides, and active ingredients such as amoxicillin, penicillin, and enrofloxacin, suggesting early-life antimicrobial exposure under professional supervision. Clusters 1 and 2 reflected intermediate or inconsistent practices, with farms reporting occasional use, uncertain protocols, or lack of defined decision criteria. Together, these patterns illustrate the wide spectrum of preventive antimicrobial strategies adopted by Brazilian dairy farms from nonuse to veterinarian-directed metaphylaxis and emphasize the need for harmonized guidelines to ensure prudent use aligned with antimicrobial stewardship goals.

DISCUSSION

This study analyzed data from 725 dairy farms located in Brazil's main milk-producing regions, grouped into 4 clusters that reflected a gradient of intensification and resource availability. This is the first comprehensive assessment of calf-rearing practices in Brazilian dairy herds, including WM feeding and antimicrobial use. Production system characteristics were embedded within the multivariate structure described by the MCA and contributed to the gradients of management intensity and antimicrobial use practices across clusters.

Cluster 3 showed the most consistent monitoring of colostrum quality and adoption of structured protocols, followed by cluster 1, which displayed partial adherence to recommended practices such as those proposed by the Dairy Calf and Heifer Association (DCHA, 2020). Cluster 2 showed limited adherence to guidelines, whereas cluster 4 relied mainly on natural suckling without standardized routines. Inadequate transfer of passive immunity markedly increases calf disease risk. Lombard et al. (2020) proposed a graded framework for passive immunity based on serum IgG levels, demonstrating that higher IgG concentrations

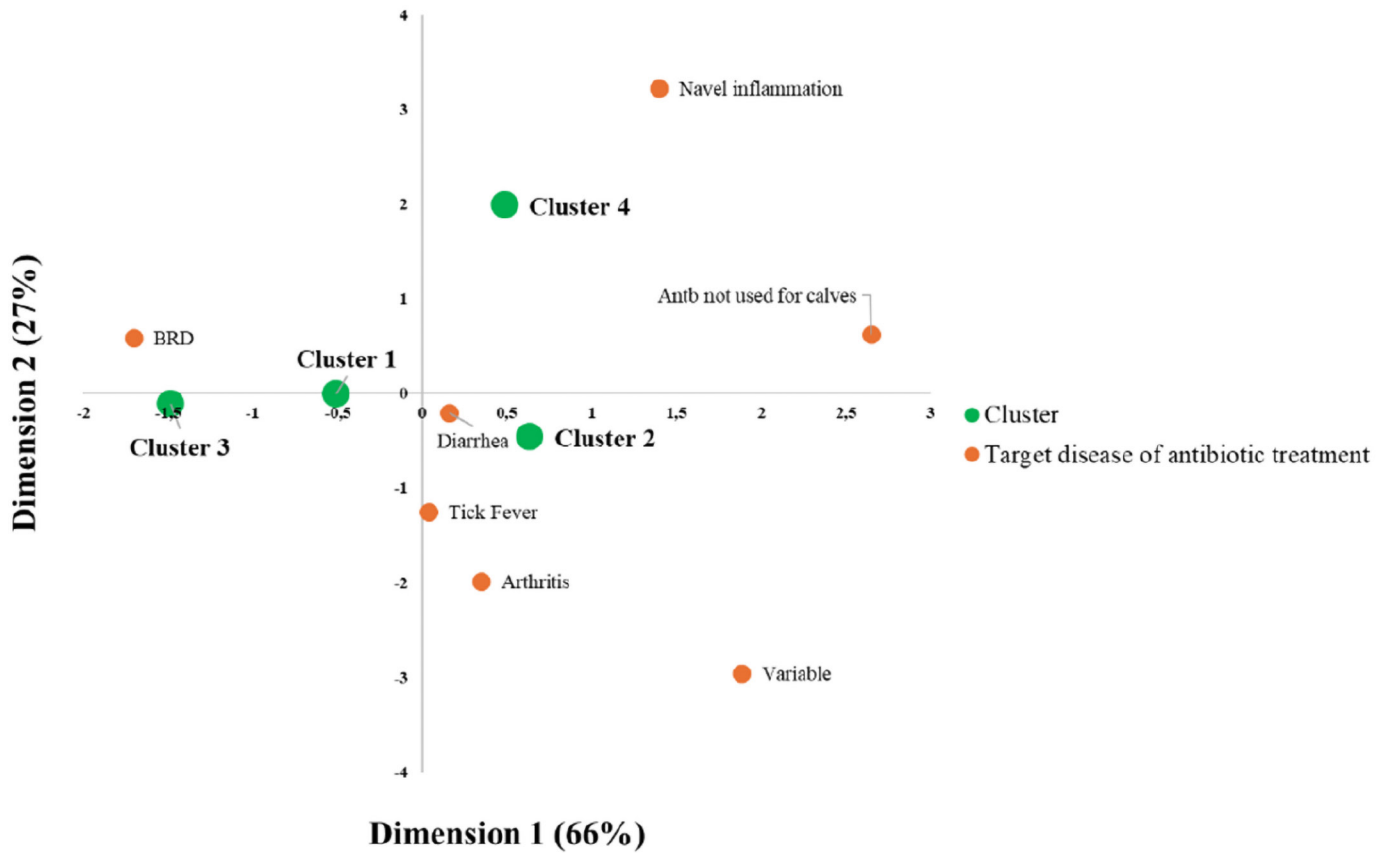


Figure 7. Multiple correspondence analysis of producers' motivation for the antimicrobial use in dairy calves in the Brazilian raising system in the preweaning period. Antb = antibiotic.

reduce morbidity and improve health outcomes. Calves with failure of passive transfer are more likely to require antimicrobial therapy, especially for respiratory and enteric diseases.

Feeding calves colostrum from cows treated with intramammary antimicrobials at dry off has raised concern. Lam et al. (2025) confirmed that colostrum from treated cows often contains antimicrobial residues. Although no association was found between these residues and the prevalence of ESBL or AmpC producing *Escherichia coli* in calf feces, the authors emphasized the importance of minimizing antimicrobial use at dry off and observing withdrawal periods.

Feeding transition milk in the first days of life was more frequent in clusters 1 and 2. Depending on the prevalence of postpartum diseases and the ability to separate milk from treated cows, transition milk may avoid early exposure to antimicrobial residues. Renaud and Steele (2025) reviewed the biological importance of transition milk, reporting benefits for intestinal development, IgG persistence, ADG, and calf survival.

Feeding management practices also varied among clusters. Cluster 3 was characterized by structured and

intensive feeding protocols, including milk replacer use, higher feeding frequency, and automated or mixed systems. Cluster 1 represented intermediate strategies, combining WM with saleable milk and using conventional delivery methods such as buckets and nipple bottles. Cluster 2 adopted more basic routines, often providing milk once daily or irregularly, in small volumes, and with minimal mechanization. Cluster 4 employed ad libitum or undefined feeding routines, with variable liquid diet compositions.

Variation in early-life feeding strategies affects calf immune development and disease resilience. Adequate nutritional planning has been associated with improved neutrophil function (Obeidat et al., 2013), faster recovery from *Cryptosporidium parvum*-induced diarrhea, better hydration, and greater ADG (Ollivett et al., 2012), as well as enhanced behavioral and physiological development (Welk et al., 2023). Intensified milk replacer programs have been linked to stronger vaccine responses and higher expression of neutrophil immune related mRNA (Hengst et al., 2012), along with elevated IFN- γ and TNF- α secretion by mononuclear cells following *Bacillus Calmette Guérin* vaccination (Foote et al., 2004).

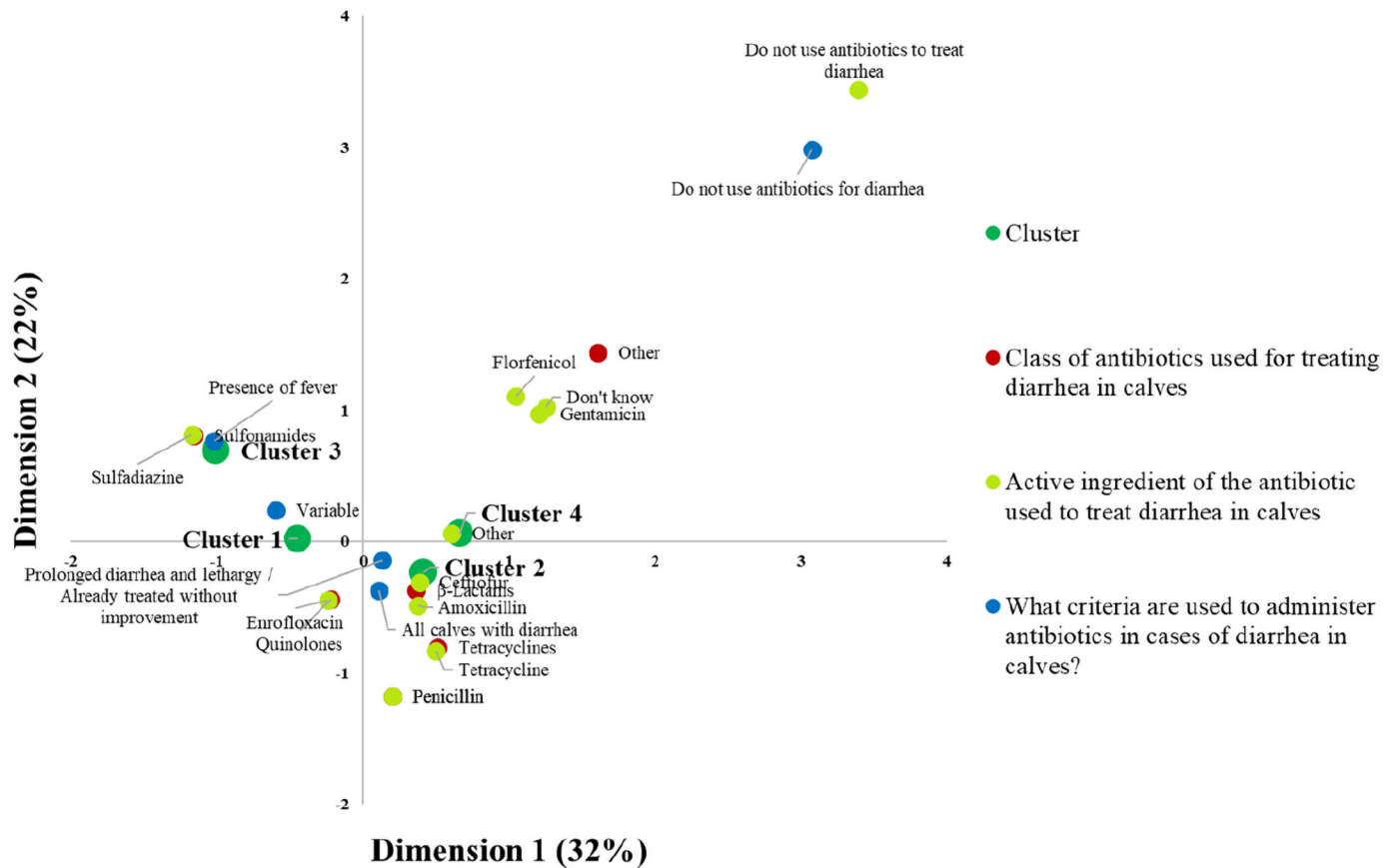


Figure 8. Multiple correspondence analysis of antimicrobial classes used for treating diarrhea in preweaning calves across clusters.

The use of WM among Brazilian dairy farms varied. Farms in cluster 1 routinely fed WM from ≤ 7 d of age and used more than 75% of the available supply. Cluster 2 farms used WM primarily due to practical constraints such as limited storage and irregular availability, typically initiating feeding between the first and fourth week of life. Clusters 1 and 2, composed of intermediate sized farms (25 to 65 lactating cows, 458 to 1,689 L/d), reported the most frequent use of WM.

Waste milk is unsuitable for human consumption when containing antimicrobial residues above regulatory thresholds. Brazilian legislation follows the maximum residue limits established by Mercosur and Codex Alimentarius, classifying such milk as unfit for industrial processing (Brasil, 2018). These legal restrictions, combined with economic pressures, explain the widespread use of WM by clusters 1 and 2, which together represented 540 of the 725 herds analyzed.

The use of WM raises antimicrobial resistance concerns because it constitutes nontherapeutic exposure to antimicrobials (Redding et al., 2019). Studies have linked WM feeding to the development of resistant gut bacteria in calves (Aust et al., 2013; Duse et al., 2015a;

Horton et al., 2016). Feeding WM to other species, such as pigs or dogs, transfers this risk across species. In response to antimicrobial resistance concerns, Brazil implemented the National Action Plan (PAN BR) in 2019 (Brasil, 2018) and later Interministerial Ordinance No. 392/2021 (Brasil, 2021), which prohibits the use of milk containing antimicrobial residues in animal feed and mandates its safe disposal.

Clusters 3 and 4 reported limited or no use of WM for calf feeding but differed in disposal practices. In cluster 3, composed of large-scale farms (average of 169 lactating cows and 5,199 L/d), milk replacers were more common, but some farms reported discharging WM into sewage systems. The dairy industry directs antibiotic-positive milk to controlled composting or licensed landfills, with contaminating farms responsible for costs. CONAMA Resolution No. 430/2011 prohibits the discharge of untreated organic effluents containing active pharmaceutical compounds (CONAMA, 2011).

Cluster 4 represented small-scale farms, often associated with natural suckling systems. Reports of WM disposal on pastures from these farms were usually at low volumes under extensive conditions. Applying ma-

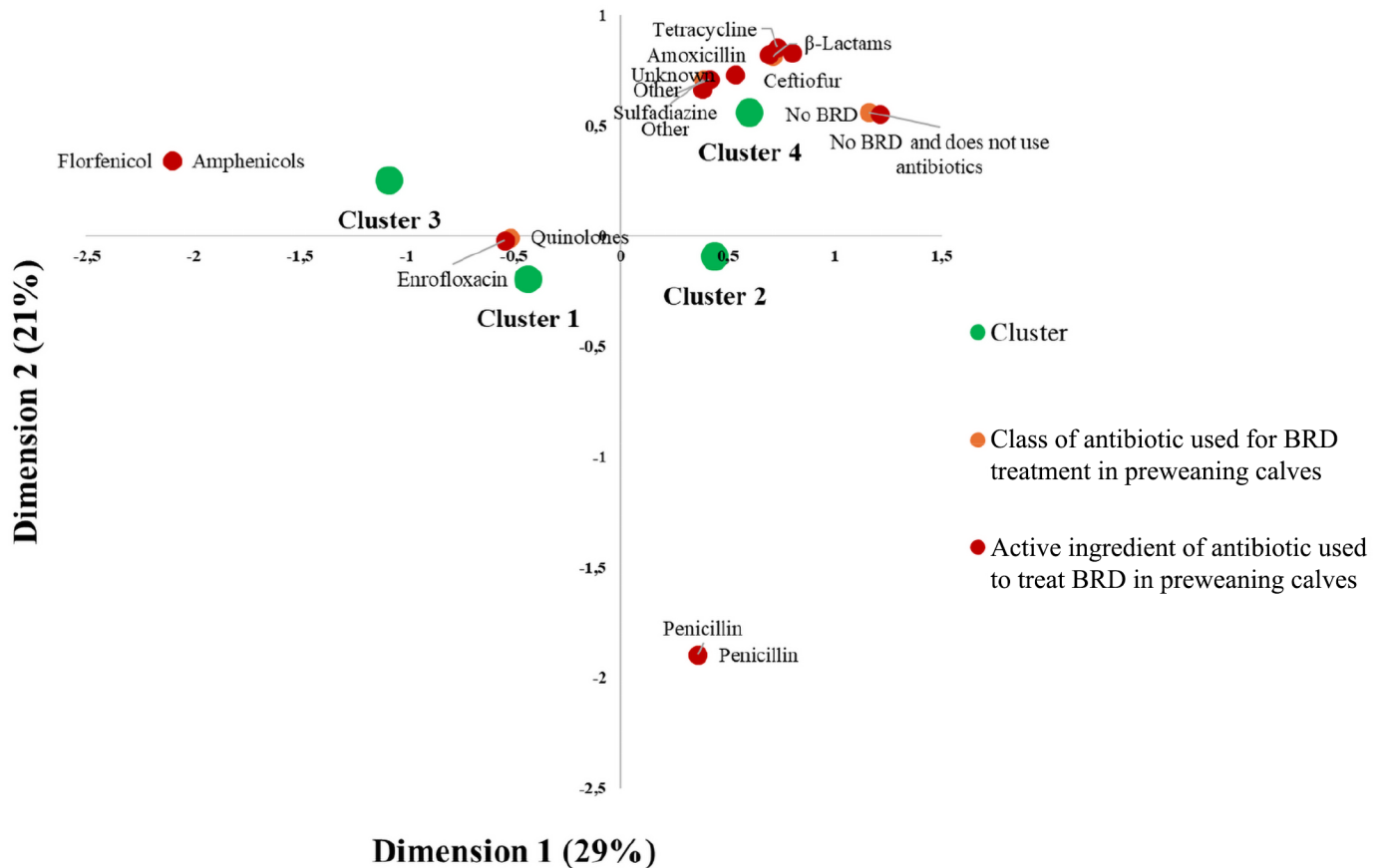


Figure 9. Multiple correspondence analysis of antimicrobial classes used for treating BRD in calves by cluster.

nure containing antimicrobial residues to pastures can promote resistant bacteria in soil (Higham et al., 2018).

Distinct patterns emerged in the motivations for antimicrobial use across clusters. Cluster 1 was moderately aligned with antimicrobial use for diarrhea and was characterized by routine treatment of diarrheic calves. Cluster 2 farms also reported antimicrobial use for diarrhea, tick fever, and arthritis. The diversity of drug classes used in this cluster, including β -lactams (e.g., ceftiofur), tetracyclines, and quinolones, indicated variable treatment strategies.

Globally, diarrhea remains the leading cause of mortality in preweaning calves (Heinrichs and Heinrichs, 2011; Windeyer et al., 2014; USDA, 2018; Azevedo et al., 2020). Protozoa, particularly *Cryptosporidium* spp., often in coinfection with rotavirus A, are the most common etiological agents of neonatal diarrhea (Izzo et al., 2011; Meganck et al., 2014). Antimicrobial therapy should be reserved for confirmed bacterial infections or cases presenting systemic signs of sepsis (Constable, 2009; Constable et al., 2017). Clinical indicators such as fever, lethargy, and the presence of blood or mucus in feces can guide therapeutic decisions (Eibl et al.,

2021; Hubbuch et al., 2021). Selective treatment protocols based on these signs reduced antimicrobial use without compromising calf health (Berge et al., 2009; Gomez et al., 2017).

According to WHO (2024), fluoroquinolones and third- to fifth-generation cephalosporins are classified as “critically important antimicrobials” (CIA) for human medicine. Their use in veterinary contexts is strongly discouraged unless no viable alternatives exist. The frequent administration of enrofloxacin in clusters 1 and 2 raises concern due to the risk of resistance.

Amoxicillin, widely used in both clusters, exhibits moderate biliary excretion and achieves effective intestinal concentrations (Berchtold, 2009). Conversely, ceftiofur and enrofloxacin are primarily excreted renally and demonstrate limited biliary elimination, resulting in poor exposure at the site of infection (Anadón et al., 1999; Foster et al., 2016a).

Bovine respiratory disease is a leading cause of morbidity and mortality in preweaning calves, with prevalence estimates ranging from 12% to 25%, depending on the production system and diagnostic method (Heinrichs and Heinrichs, 2011; Windeyer et al., 2014; USDA, 2018;

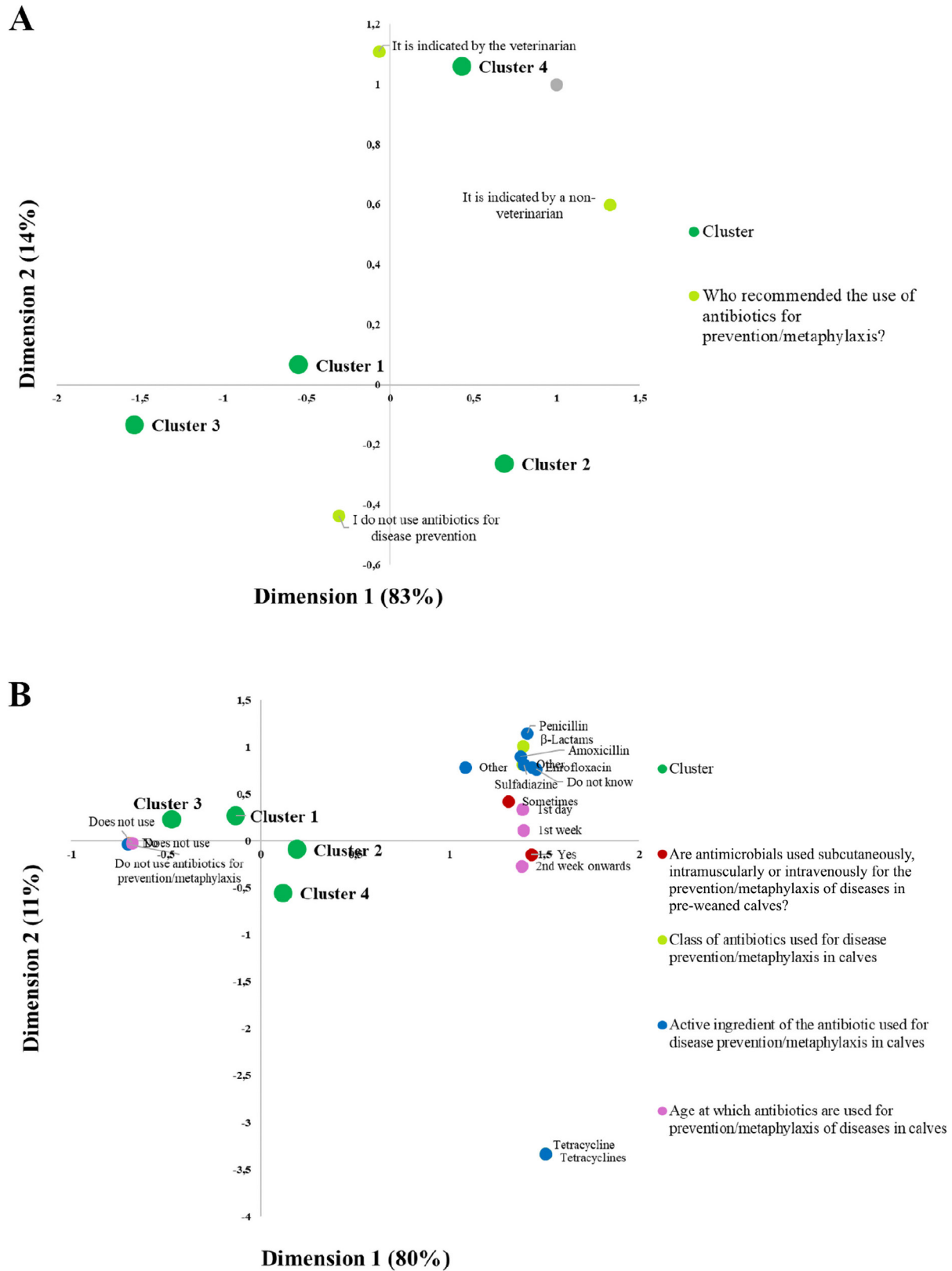


Figure 10. Multiple correspondence analyses of antimicrobial use for disease prevention/metaphylaxis in calves and its association with farm clusters: (A) Based on the source of recommendation for antimicrobial use. (B) Based on preventive antimicrobial practices, including injectable antimicrobial use, drug class and active ingredient, and age of administration.

Azevedo et al., 2020). The main bacterial agents include *Pasteurella multocida*, *Mannheimia haemolytica*, *Histophilus somni*, and *Mycoplasma bovis* (Griffin et al., 2010). Cluster 3 was associated with BRD as the primary condition motivating antimicrobial use. Clusters 3 and 1 were associated with quinolones, particularly enrofloxacin, as well as florfenicol and other amphenicols. Florfenicol is not classified as critically important by WHO (2024) and achieves therapeutic concentrations in lung tissue (de Jong et al., 2014; Foster et al., 2016b). Enrofloxacin is classified as a CIA and should be reserved for refractory cases (Scheld, 2003). For mild to moderate BRD, non-CIA alternatives such as long acting oxytetracyclines, potentiated sulfonamides, or florfenicol are effective (WHO, 2024).

Cluster 2 showed no strong association with any antimicrobial class for BRD, and cluster 4 exhibited a broad but diffuse pattern of use, including amoxicillin, ceftiofur, and tetracyclines. This cluster also showed frequent “I don’t know” responses, suggesting variable antimicrobial use.

The MCA revealed differences in how farms approached antimicrobial prevention or metaphylaxis. In cluster 4, antimicrobial decisions were reportedly guided by veterinarians. Cluster 1 represented intermediate practices, cluster 3 reported no prophylactic antimicrobial use, and cluster 2 indicated preventive use based on nonveterinarian recommendations.

Farms in clusters 2 and 4 frequently administered antimicrobials early in life, including β -lactams, sulfonamides, and fluoroquinolones, often within the first week. Martin et al. (2020) reported that tulathromycin administration within 12 h of birth did not improve clinical outcomes but altered gut microbiota development. Prophylactic antimicrobial use in healthy neonates is unjustified and potentially harmful. Fluoroquinolones and macrolides, such as enrofloxacin and tulathromycin, classified as CIA (WHO, 2024), were used across clusters. Macrolides show limited efficacy against *Escherichia coli*, the main agent of calf diarrhea (Bernal Córdoba et al., 2022). Cluster 3, composed of large structured farms, reported limited prophylactic use and therapy guided by clinical signs with florfenicol and sulfonamides.

A limitation of this study is that antimicrobial use was characterized qualitatively rather than quantitatively. Because data were obtained through self-reported questionnaires, it was not possible to estimate metrics such as the defined daily dose or defined course dose. Future research, particularly in Brazil’s major dairy belt, should aim to quantify antimicrobial use through these standardized metrics. Such data would complement the present findings by linking management typologies with actual antimicrobial consumption levels.

This study identified 4 distinct profiles of Brazilian dairy farms based on calf management and antimicrobial use practices. Antimicrobials were frequently used to treat diarrhea and respiratory disease, often without veterinary guidance and including critically important drugs such as fluoroquinolones and macrolides. Waste milk was widely offered as a liquid diet, commonly from the first week of life and without pasteurization, increasing the risk of early exposure to antimicrobial residues and resistant bacteria.

CONCLUSIONS

Larger Brazilian dairy farms, with greater access to veterinary support and technical resources, tend to adopt more structured management and antimicrobial protocols. However, BRD therapy warrants particular attention because it often includes antimicrobials classified as critically important for human health. In contrast, intermediate- and small-scale farms frequently rely on empirical approaches with limited veterinary oversight, resulting in inconsistent antimicrobial use. These findings highlight the urgent need for targeted antimicrobial stewardship programs, expanded technical support across all farm sizes, and stricter enforcement of regulations to ensure responsible use of antimicrobials and protect both animal and public health.

NOTES

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Nonstandard abbreviations used: Antb = antibiotic; BRD = bovine respiratory disease; CDA = canonical discriminant analysis; CIA = critically important antimicrobials; MCA = multiple correspondence analysis; MR = milk replacer; PCA = principal component analysis; Q1 = lower quartile; Q3 = upper quartile; WM = waste milk.

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